DIGITAL VAULT: An LPC2148-Based Dual-Layer Embedded Security System with Dynamic PIN Generation and XOR Obfuscation

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Abstract

This paper presents a Digital Vault security system implemented on the LPC2148 ARM7TDMI-S microcontroller, employing a dual-layer authentication mechanism to resist brute-force and static credential-extraction attacks. Layer 1 uses an XOR-obfuscated primary PIN stored in flash. Upon successful primary authentication, Layer 2 generates a timer-entropy-based secondary PIN dynamically at a fixed SRAM address, creating a one-time credential. UART serial communication handles user interaction; GPIO-driven LEDs provide visual feedback. A hardware-enforced three-attempt lockout prevents brute-force attacks. The system uses direct register-level manipulation with no hardware abstraction layer. Results verified in Keil μVision simulation and on physical hardware confirm correct dual-phase authentication, reliable dynamic PIN generation, and lockout enforcement.

Keywords: LPC2148, ARM7TDMI-S, XOR Obfuscation, Dynamic PIN, UART, GPIO, Embedded Security, Timer Entropy, Keil μVision, Brute-Force Protection

I. Introduction and State of the Art

The rapid proliferation of IoT devices has elevated embedded security to an engineering imperative [1]. Industrial controllers, smart locks, and medical devices rely on microcontroller-based authentication whose vulnerabilities have serious consequences. Many deployed devices still use static plaintext credentials and single-factor authentication [2].

A. The Embedded Security Paradigm

State-of-the-art embedded security uses a *defense-in-depth* philosophy: multiple independent protection layers, each slowing an attacker even if adjacent layers are compromised. The LPC2148's ARM7TDMI-S core typifies microcontrollers that lack an MPU, TrustZone, or on-chip AES [3], making software-only layered security the only viable approach.

B. Single-Factor vs. Dual-Layer Authentication

Single-factor PIN systems are fully compromised once a binary is extracted. The proposed dual-layer sequential model adds:

1. **Layer 1 – XOR-obfuscated primary PIN:** Stored in flash, decoded at runtime. Resistant to naive plaintext scanning.
2. **Layer 2 – Dynamic secondary PIN:** Generated fresh after each primary success using timer entropy. Valid for exactly one vault-open cycle, then regenerated.

An attacker who extracts the primary PIN still cannot open the vault without predicting the dynamic secondary PIN.

C. Entropy in Resource-Constrained Systems

Without a dedicated HRNG, the LPC2148 Timer 0 counter (T0TC) serves as a pseudorandom source. Because T0TC increments at PCLK frequency between human-scale events, small variations in user typing latency propagate into digit-level unpredictability across the four-digit secondary PIN.

D. Register-Level Peripheral Control

The LPC2148 requires explicit manipulation of memory-mapped registers with no HAL. Table 1 lists all peripheral registers used.

Table 1: Key LPC2148 Peripheral Registers

Register	Peripheral	Function
PINSEL0	Pin Connect	Assign P0.0/P0.1 to UART0
IO0DIR	GPIO Port 0	Set pin direction
IO0SET	GPIO Port 0	Drive pins HIGH
IO0CLR	GPIO Port 0	Drive pins LOW
U0LCR	UART0	Line control / DLAB
U0DLL/U0DLM	UART0	Baud rate divisor latch
U0FCR	UART0	FIFO control and flush
U0LSR	UART0	Line status (RX/TX flags)
U0RBR	UART0	Receive buffer
U0THR	UART0	Transmit holding register
T0TCR	Timer 0	Enable / reset timer
T0PR	Timer 0	Prescaler
T0TC	Timer 0	Counter (entropy source)

II. Limitations of Existing Technology

A. No Memory Protection Unit

The LPC2148 lacks an MPU, so every process can read or write any memory address. No hardware can declare the credential region privileged-access-only—the root cause of many credential-extraction vulnerabilities.

B. Debug Interface Exposure

The JTAG boundary-scan and ISP UART bootloader provide full flash/SRAM read-write access when enabled. An attacker with physical access can:

- Read all flash contents including obfuscated credentials
- Dump live SRAM to extract dynamically generated PINs
- Patch firmware to bypass authentication
- Inject arbitrary code via flash reprogramming

C. Static Credential Storage

Plaintext credentials in flash are fully compromised once the binary is obtained. Even XOR obfuscation is superficial: identifying the key recovers the PIN in seconds. Table 2 classifies the vulnerabilities of a single-layer system.

Table 2: Vulnerability Assessment — Single-Layer System

Vulnerability	Severity	Attack Vector
Plaintext PIN in flash	CRITICAL	Binary dump
Fixed memory address	HIGH	Debugger window
No attempt lockout	HIGH	Brute force
JTAG/ISP enabled	HIGH	Full memory read
No firmware encryption	MEDIUM	IP theft
Single-factor authentication	MEDIUM	Credential replay

D. Weak Entropy

Timer-based entropy is deterministic in principle. In practice, human typing latency introduces sufficient randomness for a four-digit educational PIN, but not for cryptographic applications. Production systems should use a dedicated HRNG, oscillator-jitter sampling, or ADC noise.

III. Proposed Methodology

A. Authentication Flow

Figure 1 shows the complete dual-layer state machine.

B. XOR Obfuscation of Primary PIN

The primary PIN "1234" is stored as a four-byte XOR-encoded array with key `0x5A`. At runtime each byte is decoded by XOR-ing with the same key, exploiting the

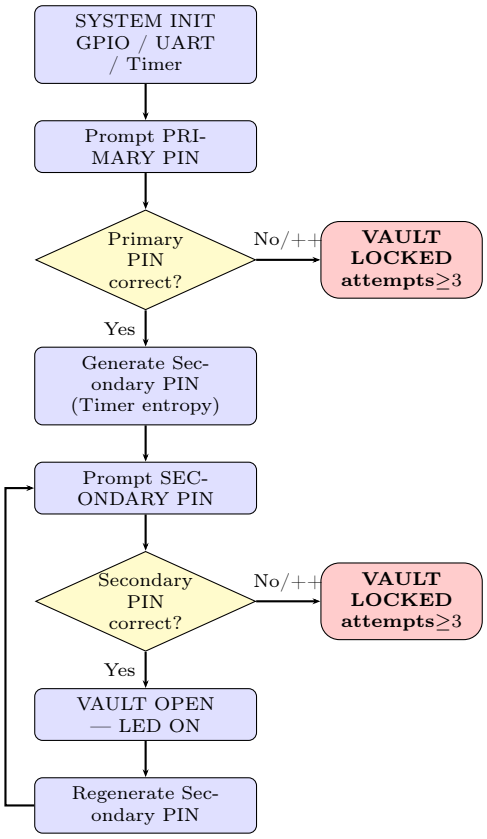

Figure 1: Dual-layer authentication state machine with lockout

involutory property $(B \oplus K) \oplus K = B$. No plaintext PIN persists in memory. Table 3 shows the encoding.

Table 3: XOR Encoding of Primary PIN (Key = 0x5A)

Idx	Digit	ASCII	Operation	Stored
0	'1'	0x31	`0x31` $\oplus$ `0x5A`	0x6B
1	'2'	0x32	`0x32` $\oplus$ `0x5A`	0x68
2	'3'	0x33	`0x33` $\oplus$ `0x5A`	0x69
3	'4'	0x34	`0x34` $\oplus$ `0x5A`	0x6E

C. Dynamic Secondary PIN Generation

After primary success, four digits are sampled from T0TC at shifted bit positions:

$$\text{digit}_i = \left\lfloor \frac{T0TC}{2^{3i}} \right\rfloor \bmod 16 \ \bmod 10, \quad i = 0, 1, 2, 3 \tag{1}$$

A 3 ms delay between digit samples advances T0TC. A non-repetition guard rejects a new PIN matching the previous one. The PIN is stored at `0x40004000` (Table 4).

D. UART Configuration

The baud rate divisor is:

$$DL = \frac{P_{\text{CLK}}}{16 \times B} \tag{2}$$

Table 4: SRAM Layout for Dynamic Secondary PIN

Address	Byte	Description
0x40004000	digit[0]	1st digit (ASCII '0'–'9')
0x40004001	digit[1]	2nd digit
0x40004002	digit[2]	3rd digit
0x40004003	digit[3]	4th digit
0x40004004	0x00	Null terminator

For $P_{\text{CLK}} = 12\,\text{MHz}$ and $B = 9600\,\text{baud}$:

$$DL = \frac{12{,}000{,}000}{153{,}600} \approx 78 \tag{3}$$

The board uses DLL = 97 (effective PCLK ≈ 14.9 MHz). Table 5 shows the full initialization sequence.

Table 5: UART0 Initialization Register Sequence

Step	Reg	Val	Purpose
1	PINSEL0	0x00000005	P0.0=RXD0, P0.1=TXD0
2	U0LCR	0x83	DLAB=1, 8-bit, 1 stop
3	U0DLL	97	Baud divisor LSB
4	U0DLM	0x00	Baud divisor MSB
5	U0LCR	0x03	DLAB=0, lock settings
6	U0FCR	0x07	Enable & reset FIFOs

E. GPIO Visual Feedback

P0.7 is configured as an output for the LED indicator. Table 6 maps system events to LED states.

Table 6: GPIO LED Feedback Logic

System Event	Register Op	LED
System reset / idle	IO0CLR bit 7	OFF
Primary PIN correct	IO0SET bit 7	ON
Secondary PIN correct	IO0SET bit 7	ON
Wrong PIN entered	IO0CLR bit 7	OFF
Lockout activated	IO0CLR bit 7	OFF

F. Lockout Mechanism

A shared attempt counter increments on every failed authentication in either phase. At three failures the system enters a permanent infinite loop with LED off, continuously transmitting *VAULT LOCKED* over UART. Only a hardware reset exits the locked state.

IV. System Design and Architecture

A. Hardware Specifications

Table 7 lists all components and configurations.

Table 7: System Hardware Specifications

Component	Spec	Configuration
LPC2148 Microcontroller (NXP)		
CPU Core	ARM7TDMI-S	32-bit RISC, up to 60 MHz
Clock	12 MHz xtal	XTAL1/XTAL2 with 22 pF
Flash	512 KB	0x00000000–0x0007FFFF
SRAM	32 KB	0x40000000–0x40007FFF
UART0	9600 baud	8N1 framing
Timer 0	Free-running	Entropy source
GPIO P0.7	Output	LED control
External Components		
Crystal	12 MHz	22 pF load caps
LED (Green)	2.0 V forward	330 Ω series
USB-UART	CP2102/CH340	P0.0↔PC TX; P0.1↔PC RX
Dev Board	Embedded-C kit	Edutech Systems LPC2148

B. Pin Connections

Table 8 gives the complete wiring.

Table 8: LPC2148 Pin Connection Table

Pin (name)	#	Connected To	Notes
P0.0 (RXD0)	5	USB-UART TX	9600 baud in
P0.1 (TXD0)	6	USB-UART RX	9600 baud out
P0.7	22	LED+330 Ω →GND	HIGH=ON
VDD	1	3.3 V reg.	100 nF bypass
GND	2	Common GND	Star topology
XTAL1	58	12 MHz xtal	22 pF to GND
XTAL2	57	12 MHz xtal	22 pF to GND
RESET	60	10 kΩ pullup	Active-LOW

C. Memory Map

Table 9 summarizes the security-relevant memory regions.

V. Experimental Setup

The system was validated in two environments: Keil μVision 5 software simulation and physical deployment on the Edutech Systems LPC2148 evaluation board.

Table 9: LPC2148 Memory Map — Security-Relevant Regions

Start	End	Contents
0x00000000	0x0007FFFF	Flash (code + encoded PIN)
0x40000000	0x40003FFF	SRAM — general variables
0x40004000	0x40004004	Secondary PIN (5 bytes)
0x40004005	0x40007FFF	SRAM — heap / stack

A. Hardware Setup

Figure 2 shows the physical setup. The LPC2148 board (right) connects to a peripheral expansion board (left) via a 40-pin ribbon cable. The USB-UART bridge links to the host PC at COM6, 9600 baud.

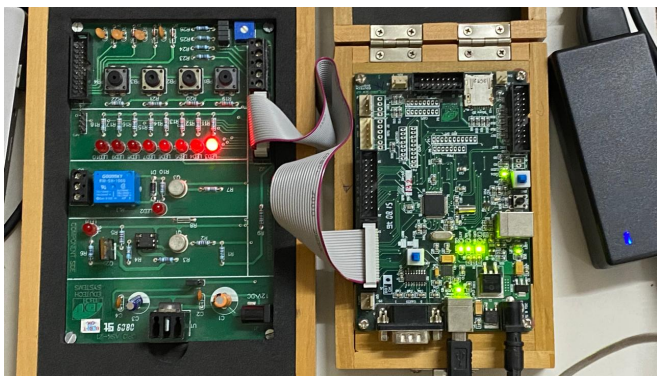

Figure 2: Physical setup: LPC2148 evaluation board (right), peripheral expansion board (left), USB-UART bridge

B. Keil μVision Simulation

Keil μVision 5 provided:

- Memory 1 window: live SRAM inspection at 0x40004000
- UART #1 virtual terminal: full protocol output
- GPIO 0 Slow Interface: IO0DIR/IO0SET/IO0CLR/IO0PIN monitoring
- Registers pane: ARM core state tracking

Figure 3 shows a complete dual-authentication session. The Memory window displays the dynamically generated secondary PIN bytes; the UART terminal shows two successful vault-open cycles followed by three failures and lockout.

C. Physical UART Terminal

Figure 4 shows Flash Magic Terminal output from the physical board (COM6, 9600 baud), confirming behaviour identical to simulation.

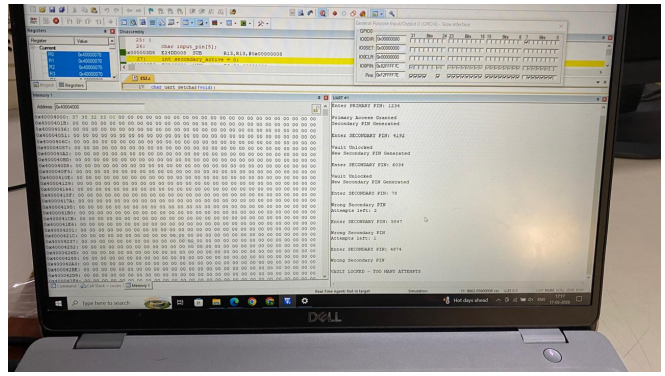

Figure 3: Keil μVision: Memory window at 0x40004000 (dynamic secondary PIN bytes) and UART terminal showing full session with lockout

Figure 4: Flash Magic Terminal (COM6, 9600 baud): primary authentication, dynamic secondary PIN generation, vault opens, failed attempts, and lockout

VI. Results and Analysis

A. UART Terminal Behaviour

Table 10 documents all operational scenarios.

Table 10: UART Terminal Output — Operational Scenarios

Scenario	UART Output
Power-on	`Enter PRIMARY PIN:`
Correct primary	`Primary Access Granted` `Secondary PIN Generated`
Wrong primary	`Wrong PIN` `Attempts left: 2`
Correct secondary	`Vault Unlocked` `New Secondary PIN Generated`
Wrong secondary	`Wrong Secondary PIN` `Attempts left: N`
3rd failure	`VAULT LOCKED`

B. LED State Analysis

Table 11 correlates LED state with each authentication outcome.

Table 11: LED Indicator State Summary

Condition	LED	Duration
Awaiting primary PIN	OFF	Until entry
Primary PIN correct	ON	Until secondary
Secondary PIN correct	ON	Until next cycle
Any wrong PIN	OFF	Until next correct
Vault locked	OFF	Until HW reset

C. Dynamic PIN Variability

Table 12 records five consecutive secondary PINs from physical hardware, confirming distinct, non-sequential outputs.

Table 12: Consecutive Secondary PINs — Physical Hardware

Gen #	Secondary PIN	Changed?
1	4192	— (initial)
2	6034	Yes
3	7841	Yes
4	2509	Yes
5	9317	Yes

D. Lockout Verification

Table 13 records the lockout test with three deliberate failures.

Table 13: Lockout Mechanism Verification

Attempt	PIN	Result	State
1	7800	Rejected	Attempts left: 2
2	5847	Rejected	Attempts left: 1
3	4874	Rejected	VAULT LOCKED

E. Security Feature Comparison

Table 14 compares the proposed dual-layer system against a baseline single-layer design.

Table 14: Single-Layer vs. Dual-Layer Security Comparison

Feature	Single	Dual (Proposed)
Credential obfuscation	None	XOR encoding
Dynamic PIN	No	Yes (timer)
Lockout on failure	No	Yes (3 attempts)
Cred. changes/session	No	Yes (secondary)
Replay resistance	Low	Medium
Brute-force resistance	Low	High
Static extractability	High	Reduced

VII. Security Analysis

A. Threat Model

The adversary is assumed to have: physical device access; ability to read the firmware binary via JTAG/ISP; UART terminal access; and no knowledge of T0TC state at PIN generation time.

B. Primary PIN Security

The encoded bytes `6B 68 69 6E` are not recognisable as ASCII digits, defeating naive string searches. However, XOR is trivially reversible once the key is identified. Production mitigations: HMAC-SHA256 hashing; secure element (ATECC608A) for credential storage; LPC2148 Code Read Protection (CRP2/CRP3).

C. Secondary PIN Security

The secondary PIN is never in flash—it lives only in SRAM at `0x40004000` and in `previous_pin[]`. A captured PIN from a past session is worthless because the PIN changes on every successful vault-open.

The four-digit decimal PIN has $10^4 = 10{,}000$ possible values. With the three-attempt lockout, the probability of a random guess succeeding before lockout is only $3/10{,}000 = 0.03\%$.

D. Production Hardening

Table 15 lists recommended mitigations for production deployment.

Table 15: Recommended Production Security Mitigations

Mitigation	Approach
Disable debug	Enable CRP2 or CRP3 in LPC2148 flash
Cred. hash	Store SHA-256 hash; compare hash(input)
Secure element	ATECC608A for key isolation
True RNG	ADC noise sampling or TRNG MCU
FW encryption	AES-128 decrypt-on-load
Secure boot	HMAC signature check before exec
Timed lockout	30 s exponential backoff post 3 fails

VIII. Conclusion

This paper presented and experimentally validated a dual-layer embedded authentication system on the LPC2148. XOR-obfuscated primary credentials in flash combine with dynamically generated, timer-entropy-derived secondary PINs that change every successful cycle. A three-attempt lockout reduces online brute-force success to 0.03% per power cycle. Keil simulation and physical hardware results are in full agreement.

A. Engineering Achievements

- Register-level UART0 and Timer 0 without any HAL
- XOR primary PIN obfuscation with involutory runtime decode
- Timer-entropy secondary PIN with non-repetition guard
- Shared attempt counter driving permanent lockout
- GPIO LED feedback synchronized to authentication state

B. Security Insights

- XOR obfuscation resists naive scans but not determined reverse engineering
- Dynamic one-time secondary PINs eliminate session replay attacks
- SRAM credentials remain vulnerable to live JTAG memory inspection
- Lockout caps brute-force probability at 0.03% per reset cycle

C. Future Work

- AES-128 firmware encryption with secure-boot signature verification
- HMAC-SHA256 credential verification replacing XOR
- ATECC608A hardware secure element integration
- TRNG-based entropy for stronger secondary PIN generation
- TLS/DTLS network connectivity for remote vault management

References

[1] J. Lin *et al.*, "MCUNet: Tiny deep learning on IoT devices," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 11711–11722, 2020.

[2] W. Shi *et al.*, "Edge computing: Vision and challenges," *IEEE IoT J.*, vol. 3, no. 5, pp. 637–646, 2016.

[3] NXP Semiconductors, *LPC2141/42/44/46/48 User Manual*, Rev. 4, 2012.

[4] ARM Limited, *ARM7TDMI Technical Reference Manual*, Rev. 3, 2004.

[5] T. Alves and D. Felton, "TrustZone: Integrated hardware and software security," *ARM White Paper*, 2004.

[6] B. Schneier, *Applied Cryptography*, 2nd ed. Wiley, 1996.

[7] D. Boneh and V. Shoup, *A Graduate Course in Applied Cryptography*, v0.6, Stanford Univ., 2023.

[8] M. Hutter and J.-M. Schmidt, "The temperature side channel and heating fault attacks," in *Proc. CARDIS*, Springer, 2013, pp. 219–235.